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SIEM Overview

Security Information and Event Management (SIEM) Overview

SIEM is a security platform that collects, centralizes, analyzes, and correlates logs from multiple security systems across the organization in order to detect security threats and generate actionable alerts.

Uses and Function

Some of the function of SIEM are:

- Data Collection/Aggregation
- Data Normalization
- Logs correlation
- Dashboarding
- Alerting

Importance of a SIEM

- It reduces the incident detection time
- It centralizes logs from different systems in a dashboard, helping SOC analysts have a holistic view of the whole security infrastructure.
- It generates alerts when an event correlation rule is triggered or when a suspicious activity that violates the security policy occurs
- It helps SOC analysts detect the specific security threat from the alert generated.
- It makes incident investigation faster.
- It correlates activity across systems

How Logs Flow in a SOC Environment

- 1. Log generation:** Logs are generated from multiples sources, including
 - Windows and Linux systems (authentication, process events)
 - Network devices (Firewalls, Proxies, Routers)
 - Security tools (IPS/IDS, email security, web security, EDR)
 - Cloud services and applications
- 2. Logs Collections and Ingestion:** Logs are forwarded to SIEM in real or near real time using:
 - Log agents
 - Syslog

- APIs or Cloud connectors

SIEM parses and normalizes the logs, so data from different sources can be in a consistent format.

3. Correlation and Detection

The SIEM applies:

- Detection rules
- Correlation logic
- Behavioral patterns

Examples:

- Multiple failed login attempts followed by a successful login
- Outbound connection to a known malicious IP address
- Suspicious Powershell execution

When the rule matches, the SIEM generates an alert.

4. Alert Creation: Alerts represent potential security incidents not a confirmed threat. Each alert contains:

- Source and destination details
- Timestamps
- Matched rule
- Severity level

Soc analysts review these alerts during triage.

What Tier-1 SOC Analysts Do with SIEM (Daily)

Tier-1 analysts mainly use SIEM for:

- Validation of alerts (True/False positive)
- Review triggered detection rules
- Investigation related logs and events
- Enriching alerts with threat intelligence
- Escalates confirmed incidents to Tier-2
- Monitoring security alerts
- Closing false positives with proper documentation

Conclusion

In real SOC environments, SIEM acts as the core detection and monitoring systems. It collects logs across the organization, correlates events and generates alerts for SOC analysts to investigate.

